

NAGAGAMI, ON, Canada

WMO: 718320

Lat: 49.7467N

Lon: 84.1636W

Elev: 264

StdP: 98.19

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-35.0	-32.0	-39.0	0.1	-34.4	-35.8	0.1	-31.8	7.6	-10.2	6.9	-10.9	1.7	300	0.646

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.3	29.0	19.6	27.2	18.4	25.5	17.7	21.4	26.4	20.2	24.8	19.1	23.3	4.6	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.9	15.0	23.6	18.6	13.9	22.5	17.4	12.9	21.6	63.8	26.7	59.5	24.6	55.5	23.4	26.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
7.3	6.4	5.7		DB	-39.3	31.9	3.3	1.6	-41.7	33.1	-43.6	34.0	-45.4	34.9	-47.8	36.0
				WB	-39.0	23.1	2.6	0.9	-40.9	23.7	-42.4	24.3	-43.9	24.8	-45.8	25.4

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	1.2	-17.8	-15.5	-8.6	-0.1	8.6	14.6	17.4	16.0	11.6	4.2	-4.3	-12.3	(6)
	DBStd	13.64	8.20	7.38	8.01	5.90	5.41	4.55	3.71	3.84	4.79	4.92	6.68	7.91	(7)
	HDD10.0	3916	862	713	578	304	93	11	2	36	193	429	692		(8)
	HDD18.3	6322	1121	947	836	553	306	127	64	94	206	439	678	951	(9)
	CDD10.0	718	0	0	1	2	50	150	229	188	86	13	1	0	(10)
	CDD18.3	82	0	0	0	0	4	17	34	22	6	0	0	0	(11)
	CDH23.3	916	0	0	0	1	76	241	339	208	50	1	0	0	(12)
	CDH26.7	208	0	0	0	0	19	59	82	41	7	0	0	0	(13)
(14) Wind	WSAvg	2.8	2.6	2.8	2.9	2.9	2.9	2.6	2.6	2.7	2.7	2.9	2.8	2.6	(14)
(15) Precipitation	PrecAvg	872	59	44	44	66	77	84	89	85	102	90	70	62	(15)
	PrecMax	1040	113	93	77	116	123	138	157	175	154	159	117	101	(16)
	PrecMin	647	31	15	9	30	40	39	31	40	47	41	38	32	(17)
	PrecStd	98	19	18	16	25	21	23	31	33	30	34	21	18	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.1	4.6	14.2	20.2	29.2	31.2	31.5	30.0	27.6	21.9	12.9	4.3	(19)
		MCWB	1.2	2.0	8.8	10.3	18.0	19.9	21.2	20.7	19.2	15.4	9.8	3.0	(20)
	2%	DB	0.4	1.6	9.5	16.0	25.3	28.2	28.9	27.7	24.6	18.2	9.4	2.3	(21)
		MCWB	-0.1	0.1	5.0	7.8	14.8	18.0	20.0	19.5	17.9	13.4	7.0	1.4	(22)
	5%	DB	-1.4	-0.8	6.1	12.7	22.2	26.2	26.9	25.7	22.1	15.2	6.7	0.7	(23)
		MCWB	-2.0	-2.0	2.6	6.2	13.3	17.0	18.8	18.8	17.2	11.7	4.8	0.0	(24)
	10%	DB	-4.3	-3.2	3.1	9.4	19.2	23.9	25.0	23.7	19.8	12.3	4.3	-0.9	(25)
		MCWB	-4.9	-4.2	0.4	4.3	11.7	16.0	18.1	17.8	15.9	9.9	3.1	-1.4	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.5	2.5	10.8	11.9	19.7	22.5	23.2	22.6	20.8	16.3	10.7	3.3	(27)
		MCDB	1.9	4.0	13.1	18.5	26.5	28.5	28.2	27.1	24.8	20.3	12.5	4.0	(28)
	2%	WB	0.0	0.2	5.5	8.5	16.5	20.2	21.6	21.0	19.2	14.2	7.3	1.4	(29)
		MCDB	0.4	1.4	8.8	14.8	22.7	25.1	26.8	25.8	22.5	17.5	8.8	2.0	(30)
	5%	WB	-1.9	-2.0	2.6	6.6	14.5	18.7	20.2	19.8	17.9	12.1	4.9	0.1	(31)
		MCDB	-1.3	-0.9	5.4	11.5	20.2	23.1	24.7	24.1	21.1	15.0	6.3	0.6	(32)
	10%	WB	-4.8	-4.2	0.6	4.7	12.6	17.2	19.0	18.5	16.2	9.8	3.1	-1.5	(33)
		MCDB	-4.3	-3.2	3.0	9.1	18.1	22.2	23.3	22.7	19.8	12.1	4.3	-1.0	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	11.2	13.1	14.0	12.5	13.3	13.1	12.3	12.0	10.9	7.9	7.6	9.2	(35)
		MCDBR	9.2	13.0	14.1	17.8	18.6	16.3	14.4	14.3	13.8	13.3	9.7	8.3	(36)
	5% WB	MCWBR	8.8	11.3	10.0	10.7	9.9	8.1	7.2	7.6	8.2	9.0	7.6	7.8	(37)
		MCWBR	9.1	11.9	12.1	15.7	15.1	12.8	12.1	12.3	11.9	12.3	9.0	8.2	(38)
(40) Clear-Sky Solar Irradiance	taub		0.269	0.261	0.293	0.346	0.360	0.364	0.391	0.369	0.354	0.311	0.277	0.265	(40)
		taud	2.245	2.311	2.294	2.319	2.402	2.414	2.359	2.429	2.463	2.557	2.507	2.310	(41)
	Ebn at Noon		774	896	918	899	899	894	862	867	844	833	785	731	(42)
		Edh at Noon	71	86	105	116	112	112	116	103	90	69	56	59	(43)
(44) All-Sky Solar Radiation	RadAvg		1.23	2.34	3.69	4.80	5.16	5.63	5.50	4.59	3.29	1.93	1.03	0.83	(44)
	RadStd		0.17	0.18	0.26	0.36	0.40	0.47	0.47	0.34	0.29	0.26	0.10	0.13	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page